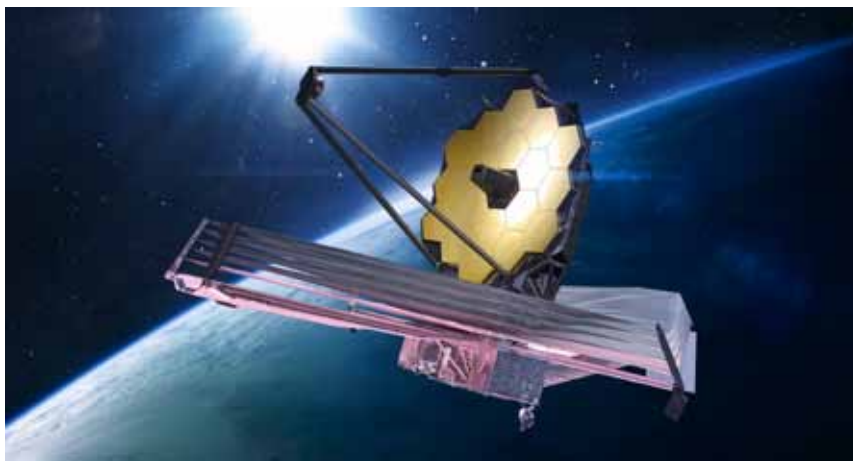




James Webb telescope exploring far galaxies and planets

This limitation has driven the development of adaptive fusion architectures that dynamically adjust each sensor's contribution based on real-time assessments of sensor health, environmental conditions, and measurement quality.

The architecture typically centres on a radiation-hardened multi-core navigation processor capable of executing extended, unscented, or particle Kalman filters. As state-vector dimensionality and measurement frequency grow, computational demand increases rapidly. To address this, modern designs incorporate dedicated hardware accelerators for matrix operations.

These accelerators, implemented as custom logic blocks within field-programmable gate arrays or as hardened intellectual-property cores in application-specific integrated circuits, deliver orders-of-magnitude improvements in throughput and power efficiency compared with software implementations on general-purpose processors.

Sensor-interface electronics add further complexity. Star trackers may provide attitude updates at five to ten hertz with arc-second accuracy, while IMUs operate at rates exceeding one kilohertz but accumulate drift. Interface circuitry therefore provides microsecond-precision time-stamping, typically synchronised to a central spacecraft clock derived from an ultra-stable

oscillator such as an oven-controlled crystal oscillator or, for the highest-precision missions, a mercury-ion atomic clock.

Adaptive multi-source data fusion

Sensor fusion combines complementary measurements to provide accurate and robust navigation while accounting for heterogeneity, non-Gaussian noise, and varying reliability. Classical approaches include Kalman filters, extended Kalman filters, and unscented Kalman filters, while particle filters and interacting multiple-model methods address multimodal or non-Gaussian conditions. Federated filtering scales effectively for distributed systems.

Adaptive fusion dynamically adjusts weights and models based on sensor health and context, down-weighting faulty inputs while relying more heavily on reliable sources. Covariances are estimated online, FDIR-derived health factors are applied, and estimator topologies may switch as conditions change.

Recent advances introduce meta-estimators with risk metrics and adversarial robustness that are now maturing for flight deployment.

A common spacecraft architecture employs a layered estimator: hardware observers generate low-latency residuals; local processors perform preconditioning and time-stamping; a mid-level engine

operates a filter bank to achieve consensus; and a mission-level manager oversees trajectories and operating modes. This structure enables graceful degradation and fault-specific recovery policies, ensuring navigation continuity even under adverse conditions.

AutoNav and deep space

The JPL AutoNav experiment on NASA's Deep Space 1 mission established the feasibility of optical autonomous navigation for interplanetary missions. AutoNav used onboard imaging to localise asteroids against the starfield and autonomously update trajectory estimates, reducing ground workload and enabling tighter proximity operations.

Its success demonstrated that image processing, feature extraction, and filter-based state estimation could be performed on board, enabling the development of today's more advanced autonomous navigation suites. AutoNav's historical significance is not merely archival, its core design principles, onboard perception, onboard estimation, and autonomy-based mission assurance, remain relevant and continue to shape modern systems.

SEXTANT and X-ray pulsar navigation

The SEXTANT project used NASA's NICER payload on the International Space Station to demonstrate X-ray pulsar navigation (XNAV). XNAV leverages the predictable timing of millisecond pulsars as natural beacons.

SEXTANT showed that pulse time-of-arrival measurements can yield position fixes independent of ground stations. While not yet a replacement for the deep space network (DSN) in routine operations, pulsar navigation provides a compelling backup and a practical autonomy pathway for missions far from Earth, where DSN access becomes sparse.

The demonstration validated the hardware and algorithms required for pulsar detection and time-tagging,